

Cover Letter - Transportation Research Part E

31 August 2026

Dear Editors of *Transportation Research Part E: Logistics and Transportation Review*,

Please consider my manuscript, “Temporal Evaluation, Calibrated Risk, and Conformal Delay-Severity Prediction in Pharmaceutical Logistics,” for publication as a research article.

The study addresses a logistics-evaluation problem that arises when international shipment-delay models are trained and assessed on historical records whose outcomes mature over time. Using USAID SCMS pharmaceutical and health-commodity shipments, we compare standard random cross-validation with a leakage-aware expanding temporal protocol incorporating a 90-day post-delivery embargo. The paper then connects calibrated delay-risk probabilities to conditional delay-severity modeling, conformalized quantile uncertainty, and capacity-constrained shipment prioritization.

The primary empirical result is that random cross-validation overestimates PR-AUC relative to temporally ordered evaluation across all five tested classifier families, with relative inflation ranging from 26.2% to 99.7%. We also report negative and mixed findings transparently: Random Forest outperforms the deployment-aligned CatBoost model on several secondary benchmark metrics; a conditional median is stronger than the quantile model on point MAE; and uncertainty-aware prioritization is not universally superior across inspection capacities.

We believe the manuscript fits TR-E because its scientific focus is logistics reliability, shipment lead-time uncertainty, risk/disruption analytics, and decision support under constrained operational capacity. The contribution is methodological integration and evaluation discipline rather than a new machine-learning algorithm.

The manuscript uses a public historical dataset, reports frozen provenance hashes and a reproducible temporal protocol, and clearly labels all operational prioritization results as simulated scenarios.

[FINAL AUTHOR CONFIRMATION REQUIRED BEFORE SUBMISSION: This manuscript is original, has not been published previously, and is not under consideration by another journal.]

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